Example 29.3

Maxwell's Equations for Electromagnetic Fields.
Define the following quantities

- electric charge density $\rho(\mathbf{r}, t)$
- electric current $\boldsymbol{J}(\mathbf{r}, t)$
- vector field for magnetic force $\boldsymbol{B}(\mathbf{r}, t)$
- vector field for electric force $\boldsymbol{E}(\mathbf{r}, t)$
- permittivity of free space ε_0
- permeability of free space μ_0.

In S.I. units, Maxwell's equations can be written as

(a) $\boldsymbol{\nabla} \times \boldsymbol{E} = -\dfrac{\partial \boldsymbol{B}}{\partial t}$ (Faraday's Law).
(If $\boldsymbol{B}$ changes with time, then an electric field is produced)

(b) $\boldsymbol{\nabla} \cdot \boldsymbol{E} = \dfrac{\rho}{\varepsilon_0}$ (Gauss' Law).
(Charges present make $\boldsymbol{\nabla} \cdot \boldsymbol{E}$ nonzero.)

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(c) $\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \varepsilon_0 \mu_0 \dfrac{\partial \boldsymbol{E}}{\partial t}$ (Ampere's Law).
(If $\boldsymbol{\nabla} \times \boldsymbol{B} \neq \boldsymbol{0}$ then there are currents or $\boldsymbol{E}$ changes with time.)

(d) $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$.
($\boldsymbol{B}$ is always incompressible. There are no magnetic sources.)

Derive some consequences of Maxwell's equations.

Solution:

1. Assume $\boldsymbol{B}$ is constant in time.
   Eqn (a) gives $\boldsymbol{\nabla} \times \boldsymbol{E} = \boldsymbol{0}$. So $\boldsymbol{E}$ is a conservative field and we have

$$\boldsymbol{E} = \boldsymbol{\nabla} \phi$$

   for some scalar-valued function ϕ. ϕ is called the electric potential.

Solution:

2. Assume $\boldsymbol{E}$ is constant in time.
   Eqn (c) gives $\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J}$

   So if there is an electric current there must be a magnetic field.
3. If there are no currents or charges (so $\mathbf{J} = \mathbf{0}$ and $\rho = 0$)
   - Eqn (b) gives $\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0$.
   - Eqn (c) gives $\boldsymbol{\nabla} \times \boldsymbol{B} = \varepsilon_0 \mu_0 \dfrac{\partial \boldsymbol{E}}{\partial t}$
     Differentiate with respect to time

$$\boldsymbol{\nabla} \times \frac{\partial \boldsymbol{B}}{\partial t} = \varepsilon_0 \mu_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2}$$

   - Taking curl of Eqn (a) gives

$$\begin{aligned} \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) &= -\boldsymbol{\nabla} \times \left(\frac{\partial \boldsymbol{B}}{\partial t} \right) \\ &= -\varepsilon_0 \mu_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} \end{aligned}$$

Solution:

- Using vector identity 15, we get

$$\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{E}) - \nabla^2\boldsymbol{E} = -\varepsilon_0\mu_0\frac{\partial^2\boldsymbol{E}}{\partial t^2}$$

Since $\boldsymbol{\nabla}\cdot\boldsymbol{E} = 0$

$$\Rightarrow \nabla^2\boldsymbol{E} = \varepsilon_0\mu_0\frac{\partial^2\boldsymbol{E}}{\partial t^2}$$

Note

This is the wave equation in a vacuum. The wave equation can be written in the form

$$\nabla^2 \boldsymbol{E} = \frac{1}{c^2}\frac{\partial^2 \boldsymbol{E}}{\partial t^2}$$

where c is the speed of the wave.
Note that the permittivity ε_0 of free space is

$$8.85418782 \times 10^{-12}\ \mathrm{m}^{-3}\ \mathrm{kg}^{-1}\ \mathrm{s}^4\ \mathrm{A}^2$$

and the permeability μ_0 of free space is

$$1.25663706 \times 10^{-6}\ \mathrm{m}\ \mathrm{kg}\ \mathrm{s}^{-2}\ \mathrm{A}^{-2}.$$

The electromagnetic waves travel with speed

$$\frac{1}{\sqrt{\varepsilon_0 \mu_0}} \approx 300,000\ \mathrm{km/sec}.$$

This was predicted by Maxwell 20 years before experimentally observed by Hertz.